WORKED SOLUTION

STEP III 2003, Question 9

Mechanics • SHM

STEP III 2003, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q9

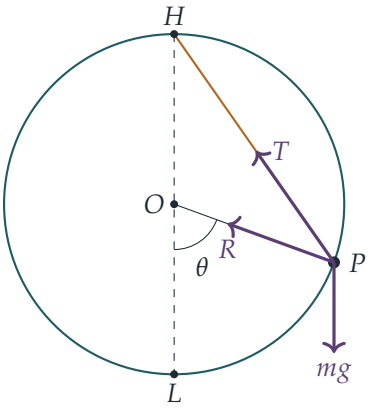
The key idea

The wire is smooth, so its reaction is always along the radius and does no work. Everything therefore happens in the *tangential* direction: resolve the tension and the weight along the tangent and you have a single equation for $\ddot{\theta}$ that gives both the equilibrium and the small oscillations. The geometry of the isosceles triangle OHP supplies all the angles.

Solution

Setting up

Model P as a particle on a smooth wire, so the reaction R from the wire acts along PO. The string is light and obeys Hooke's law. Take $0 < \theta < \pi$, so P is on one side of the vertical diameter LH.



Since $OH = OP = a$, triangle OHP is isosceles with $\angle HOP = \pi - \theta$, so its base angles

are

$$\angle OHP = \angle OPH = \tfrac{1}{2}\theta, \qquad HP = 2a\cos\tfrac{1}{2}\theta.$$

The string is taut when $HP > a$, that is, when $\cos\frac{1}{2}\theta > \frac{1}{2}$, and then Hooke's law gives

$$T = \frac{\alpha mg}{a}(2a\cos\tfrac{1}{2}\theta - a) = \alpha mg(2\cos\tfrac{1}{2}\theta - 1).$$

Now resolve along the tangent at P, in the direction of increasing θ (towards H). The string makes angle $\frac{1}{2}\theta$ with PO, so it makes angle $\frac{1}{2}\pi - \frac{1}{2}\theta$ with this tangent, and its tangential component is $T\sin\frac{1}{2}\theta$. The weight has tangential component $-mg\sin\theta$, and R has none. With tangential acceleration $a\ddot{\theta}$,

$$\begin{aligned} ma\ddot{\theta} &= T\sin\tfrac{1}{2}\theta - mg\sin\theta \\ &= mg\left[\alpha(2\cos\tfrac{1}{2}\theta - 1)\sin\tfrac{1}{2}\theta - 2\sin\tfrac{1}{2}\theta\cos\tfrac{1}{2}\theta\right], \end{aligned}$$

so, while the string is taut,

$$\ddot{\theta} = \frac{g}{a}\sin\tfrac{1}{2}\theta\left[2(\alpha-1)\cos\tfrac{1}{2}\theta - \alpha\right]. \qquad (*)$$

(The same equation comes from differentiating the energy $\frac{1}{2}ma^2\dot{\theta}^2 - mga\cos\theta + \frac{1}{2}\alpha mga(2\cos\frac{1}{2}\theta - 1)^2$.)

The equilibrium

In equilibrium $\ddot{\theta} = 0$. For $0 < \theta < \pi$ we have $\sin\frac{1}{2}\theta \neq 0$, so we need

$$\cos\tfrac{1}{2}\theta = \frac{\alpha}{2(\alpha-1)},$$

which is allowed since $\alpha > 1$. For this to give $0 < \theta < \pi$ we need $0 < \cos\frac{1}{2}\theta < 1$. The value is positive, and

$$\frac{\alpha}{2(\alpha-1)} < 1 \iff \alpha < 2\alpha - 2 \iff \alpha > 2.$$

We also need the string to be taut there, as $(*)$ assumes: $\dfrac{\alpha}{2(\alpha-1)} > \dfrac{1}{2} \iff \alpha > \alpha - 1$, which is always true. So if $\alpha > 2$ there is an equilibrium position with $0 < \theta < \pi$, as required.

Ask yourself What if the string were slack at the equilibrium?

Then only gravity would have a tangential component, and $-mg\sin\theta = 0$ has no solution with $0 < \theta < \pi$. That's why it matters that the equilibrium we found has $\cos\frac{1}{2}\theta > \frac{1}{2}$ automatically.

Small oscillations when $\alpha = 2 + \sqrt{2}$

Now

$$\cos\tfrac{1}{2}\theta = \frac{2+\sqrt{2}}{2(1+\sqrt{2})} = \frac{\sqrt{2}(\sqrt{2}+1)}{2(\sqrt{2}+1)} = \frac{1}{\sqrt{2}},$$

so the equilibrium is at $\frac{1}{2}\theta = \frac{1}{4}\pi$, that is, $\theta = \frac{1}{2}\pi$, with P level with O. Put $\theta = \frac{1}{2}\pi + \phi$, so $\ddot{\theta} = \ddot{\phi}$. Write $(*)$ as $\ddot{\theta} = \frac{g}{a} f(\theta)$, where $f(\frac{1}{2}\pi) = 0$. For small ϕ,

$$f(\tfrac{1}{2}\pi + \phi) \approx f'(\tfrac{1}{2}\pi)\,\phi.$$

Because the square bracket in $(*)$ vanishes at $\theta = \frac{1}{2}\pi$, the product rule leaves only one term:

$$f'(\tfrac{1}{2}\pi) = \sin\tfrac{1}{4}\pi \cdot \left[-2(\alpha-1)\cdot\tfrac{1}{2}\sin\tfrac{1}{4}\pi\right] = -(\alpha-1)\sin^2\tfrac{1}{4}\pi = -\frac{\alpha-1}{2} = -\frac{\sqrt{2}+1}{2}.$$

Therefore

$$\ddot{\phi} \approx -\frac{g(\sqrt{2}+1)}{2a}\,\phi$$

for small ϕ, as required.

Final part

The coefficient of ϕ is negative, so a small displacement makes the particle oscillate about $\theta = \frac{1}{2}\pi$ in (approximately) simple harmonic motion: the equilibrium is stable. The angular frequency is $\omega = \sqrt{g(\sqrt{2}+1)/(2a)}$, so the period is

$$\frac{2\pi}{\omega} = 2\pi\sqrt{\frac{2a}{(\sqrt{2}+1)g}} = 2\pi\sqrt{\frac{2(\sqrt{2}-1)a}{g}},$$

and the amplitude stays small, because energy is conserved (the wire is smooth and the string is elastic, so nothing dissipates energy).

Checks

- At $\alpha = 2$ the equilibrium formula gives $\cos\frac{1}{2}\theta = 1$, i.e. $\theta = 0$: the equilibrium merges with the lowest point, which fits the condition $\alpha > 2$.
- As $\alpha \to \infty$, $\cos\frac{1}{2}\theta \to \frac{1}{2}$, so $\theta \to \frac{2}{3}\pi$ and $HP \to a$: a very stiff string barely stretches.
- Stepping the full equation $(*)$ numerically from $\phi = 0.01$ with $a = 1.3\,\mathrm{m}$ and $g = 9.81\,\mathrm{m\,s^{-2}}$ gives a period of about $2.0819\,\mathrm{s}$, matching $2\pi\sqrt{2(\sqrt{2}-1)a/g} \approx 2.0818\,\mathrm{s}$.

Common slips

- Using $HP = 2a\sin\frac{1}{2}\theta$: that's the chord LP, not HP. The angle at O in triangle OHP is $\pi - \theta$.

- Forgetting to check that the string is taut at the equilibrium, since a slack string has no tension.
- Resolving radially instead of tangentially: the radial equation brings in the unknown reaction R and tells you nothing about $\ddot{\theta}$.

Beyond STEP

Linearising about an equilibrium, $\ddot{\phi} \approx f'(\theta_0)\,\phi$, is the standard first step in studying stability of any dynamical system: the sign of $f'(\theta_0)$ (equivalently of $V''(\theta_0)$ for the potential energy) decides between oscillation and runaway. In a first-year course this grows into the phase-plane analysis of nonlinear ODEs. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats small oscillations about stable equilibria via potential energy very clearly.

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